ii) Equivalent internal resistance r_{eq} = nr
$= 4 \times 0.1 = 0.4\ \Omega$

iii) Total current $I = \frac{n\xi}{R+nr}$

$= \frac{4\times 9}{10+(4\times 0.1)}$

$= \frac{4\times 9}{10+0.4} = \frac{36}{10.4}$

I = 3.46 A

iv) Potential difference across external resistance V = IR = 3.46 × 10 = 34.6 V. The remaining 1.4 V is dropped across the internal resistance of cells.

v) Potential difference across each cell $\frac{V}{n} = \frac{34.6}{4} = 8.65V$

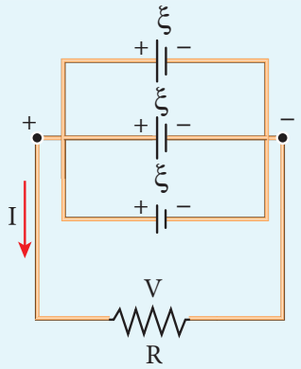
2.4.4 Cells in parallel

In parallel connection all the positive terminals of the cells are connected to one point and all the negative terminals to a second point. These two points form the positive and negative terminals of the battery.

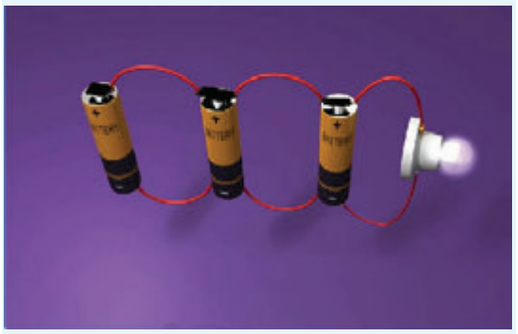
Let n cells be connected in parallel between the points A and B and a resistance R is connected between the points A and B as shown in Figure 2.22. Let ξ be the emf and r the internal resistance of each cell.

The equivalent internal resistance of the battery is $\frac{1}{r_{eq}} = \frac{1}{r} + \frac{1}{r} + ...\frac{1}{r}(nterms) = \frac{n}{r}$. So $r_{eq} = \frac{r}{n}$ and the total resistance in the circuit $= R + \frac{r}{n}$. The total emf is the potential difference between the points A and B, which is equal to ξ. The current in the circuit is given by

$$I = \frac{\xi}{\frac{r}{n} + R}$$



cells in parallel (Circuit diagram)



Cells in parallel (Schematic diagram)

Figure 2.22 Cells in parallel

$$I = \frac{n\xi}{r+nR} \qquad (2.42)$$

Case (a) If r >> R, $I = \frac{n\xi}{r} = nI_1$ (2.43)

where I_1 is the current due to a single cell and is equal to $\frac{\xi}{r}$ when R is negligible. Thus, the current through the external resistance due to the whole battery is n times the current due to a single cell.

Case (b) If r<<R, $I = \frac{\xi}{R}$ (2.44)



When the car engine is started with headlights turned on, they sometimes become dim. This is due to the internal resistance of the car battery.